

Interactive Parrot Robot

with Face Tracking, Emotion Recognition and Voice Repetition

Overview

Freddy is an interactive parrot-inspired animatronic robot that runs locally and offline on a **Raspberry Pi 5**. Designed in Fusion 360, built using 3D-printed parts, and actuated with **MG90S** and **MG996R** servos, Freddy aims to provide engaging entertainment and educational interaction without relying on cloud services.



Technical Features

Freddy was coded in **Python**, using open-source libraries such as **MediaPipe** for face detection, **HSEmotion** for emotion recognition, and **Whisper** for speech-to-text transcription. It also uses a custom **Piper** voice for text-to-speech, trained on **ElevenLabs**-generated audio. Additionally, the wake-word model was generated using **Porcupine**.

How to Interact with Freddy

- 1. Robot Mode (Default):** Freddy tracks the user's face using two **Raspberry Pi Camera Module 2** cameras mounted in its eyes and recognises seven emotions. It then responds with eyelid and wing movement, speech, and LED ring colour changes.
- 2. Parrot Mode:** Freddy listens for a wake word, transcribes spoken input, translates it into English, and repeats it aloud. Simply say "**Hey Freddy**" to activate **Parrot Mode**.



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Buck Converter, Level Shifter, MG90S
and PCA9685 Images from Amazon.com
Raspberry Pi 5 and Raspberry Pi Camera
Module 2 Images from ThePiHut.com